The Balance-Sample Size Frontier in Matching Methods for Causal Inference[1]

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(Talk at the University of Michigan, 1/24/2014)

[1] Joint work with Christopher Lucas and Richard Nielsen
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  - Problem: Propensity score matching increases imbalance!
  - Solution: Not an issue with other methods or our approach

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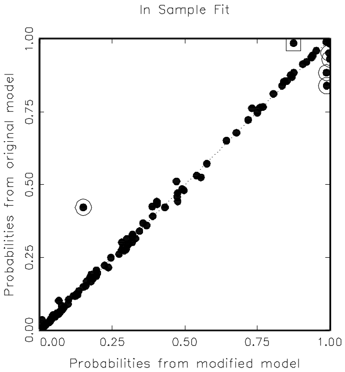
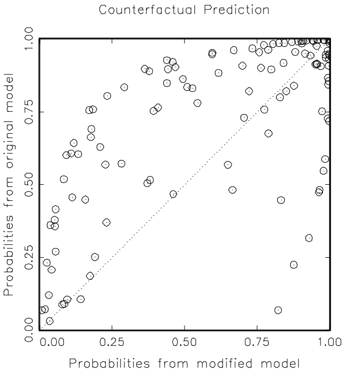
- Data analysis: Logit model
- The question: How *model dependent* are the results?

Two Logit Models, Apparently Similar Results

	Original "Interactive" Model			Modified Model		
Variables	Coeff	SE	P-val	Coeff	SE	P-val
Wartype	−1.742	.609	.004	−1.666	.606	.006
Logdead	−.445	.126	.000	−.437	.125	.000
Wardur	.006	.006	.258	.006	.006	.342
Factnum	−1.259	.703	.073	−1.045	.899	.245
Factnum2	.062	.065	.346	.032	.104	.756
Trnsfcap	.004	.002	.010	.004	.002	.017
Develop	.001	.000	.065	.001	.000	.068
Exp	−6.016	3.071	.050	−6.215	3.065	.043
Decade	−.299	.169	.077	−0.284	.169	.093
Treaty	2.124	.821	.010	2.126	.802	.008
UNOP4	3.135	1.091	.004	.262	1.392	.851
Wardur*UNOP4	—	—	—	.037	.011	.001
Constant	8.609	2.157	0.000	7.978	2.350	.000
N		122			122	
Log-likelihood		-45.649			-44.902	
Pseudo R^2		.423			.433	

Doyle and Sambanis: Model Dependence





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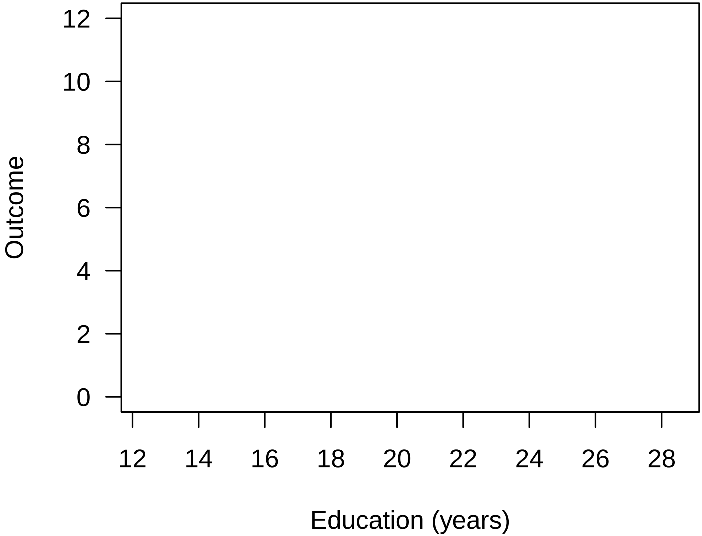

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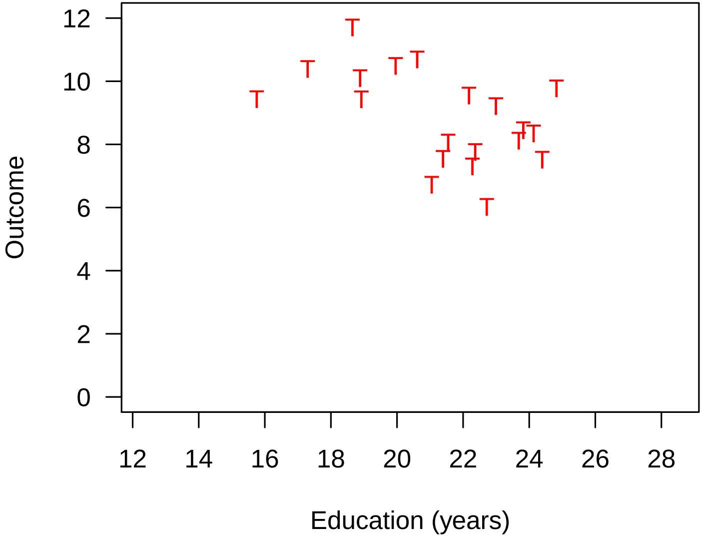

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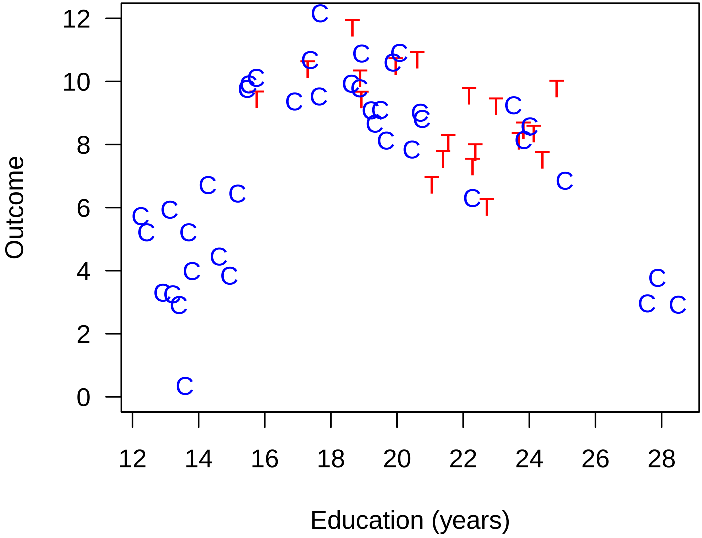

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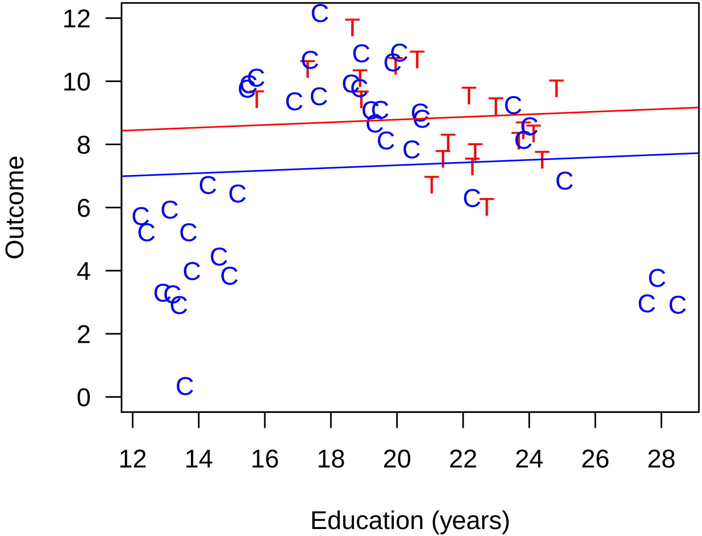

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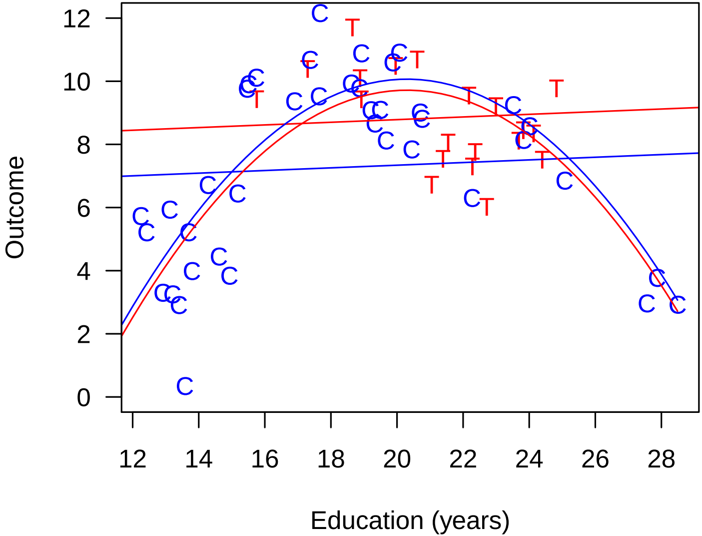

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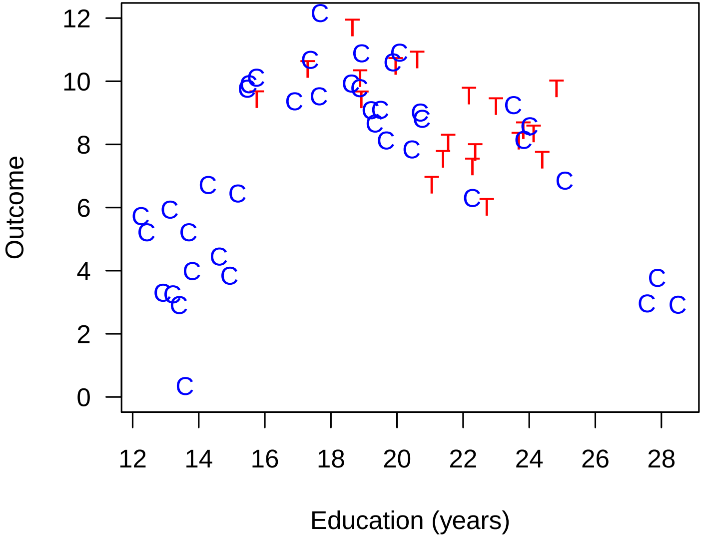

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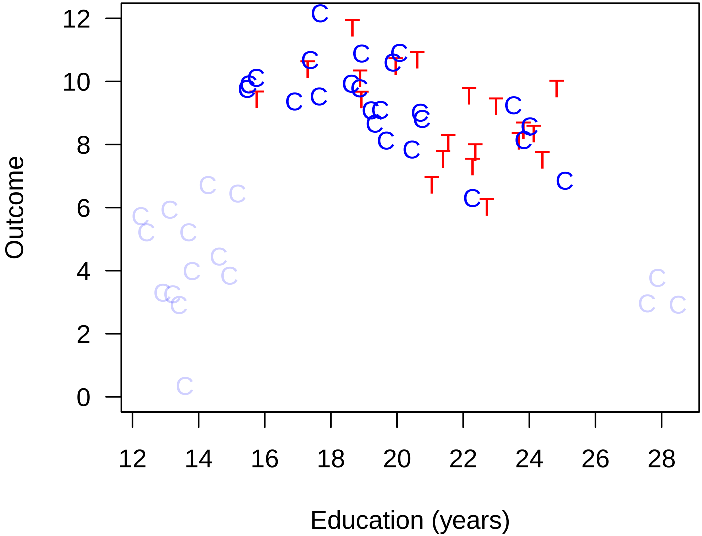

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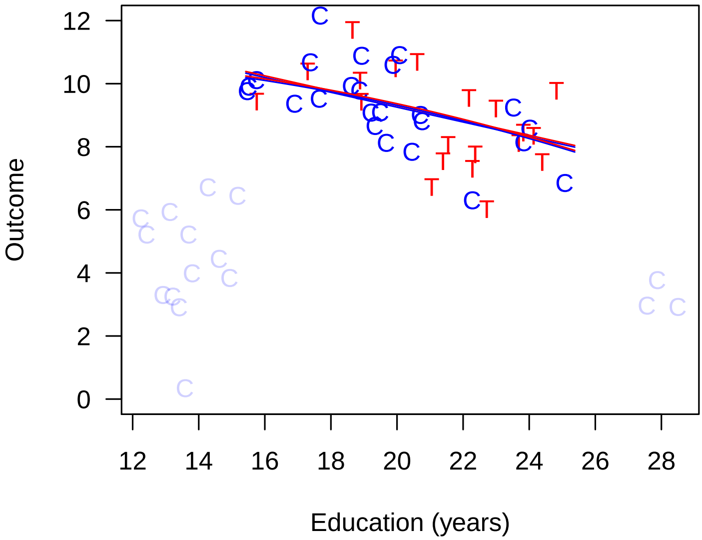

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Matching reduces model dependence, bias, and variance

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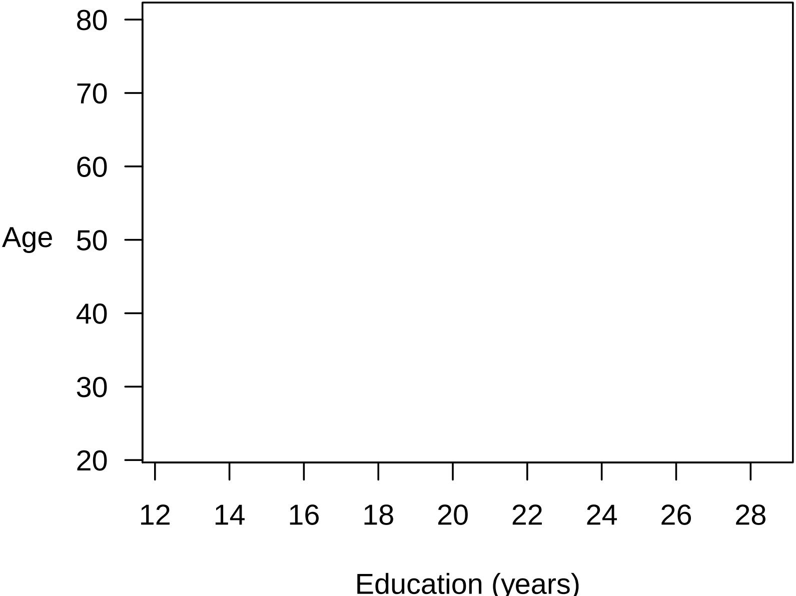

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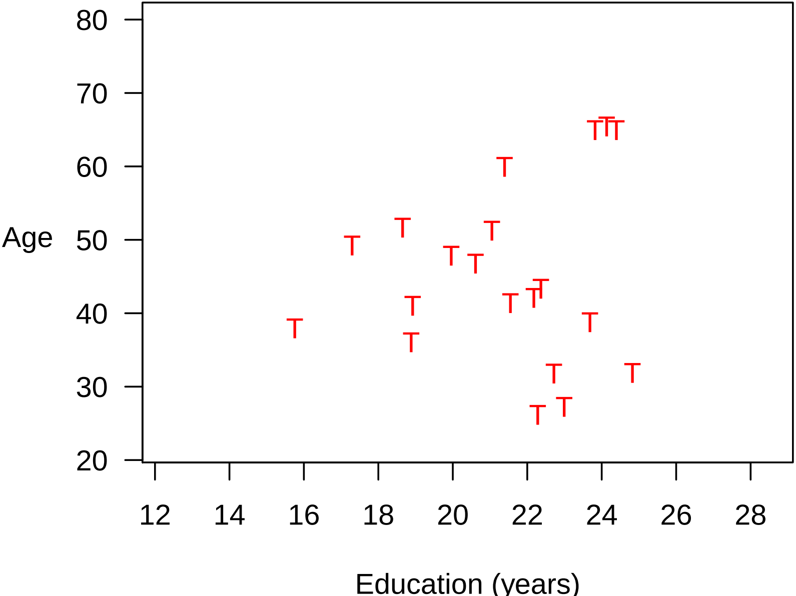

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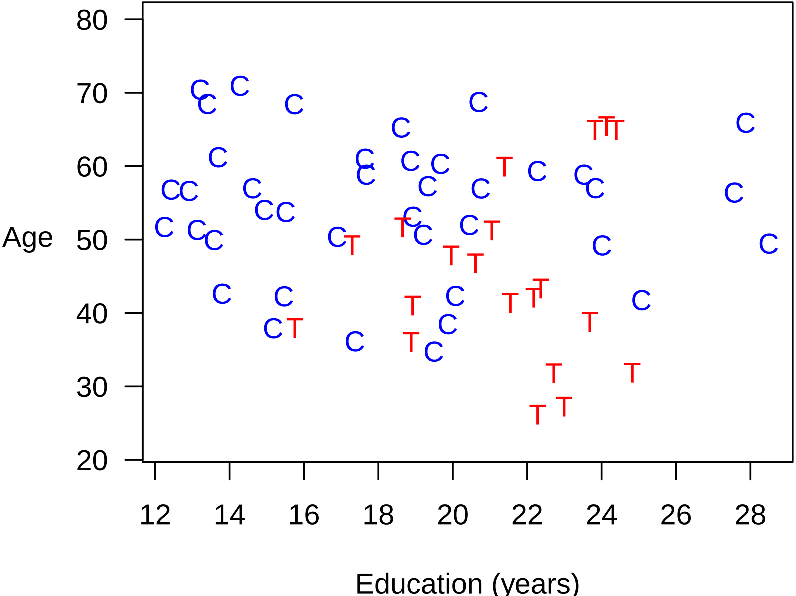

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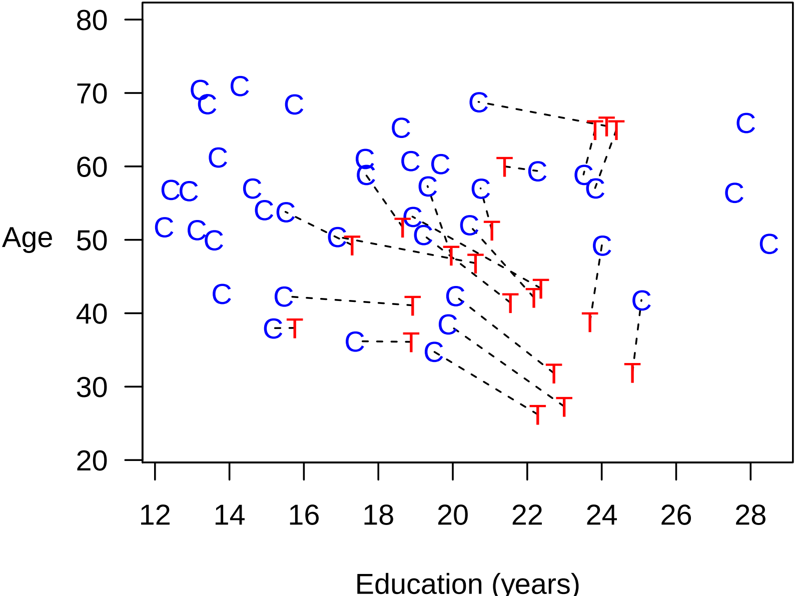

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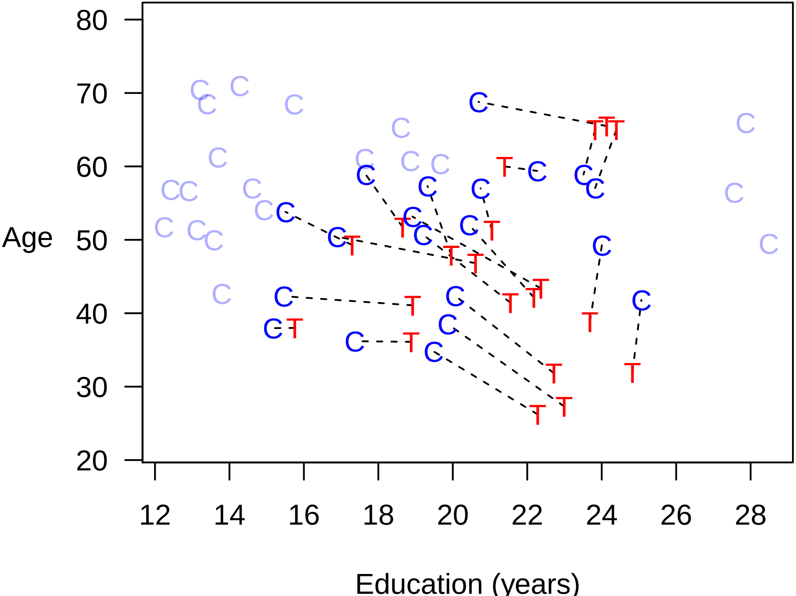

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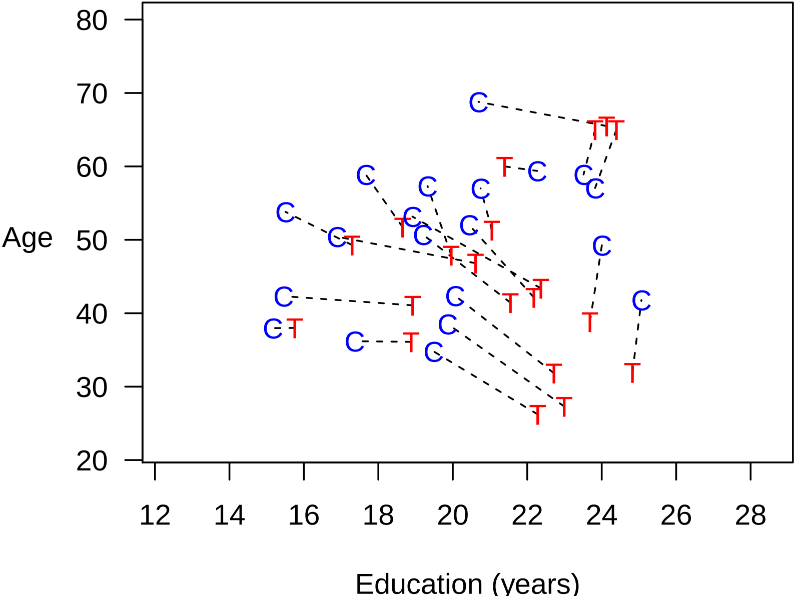

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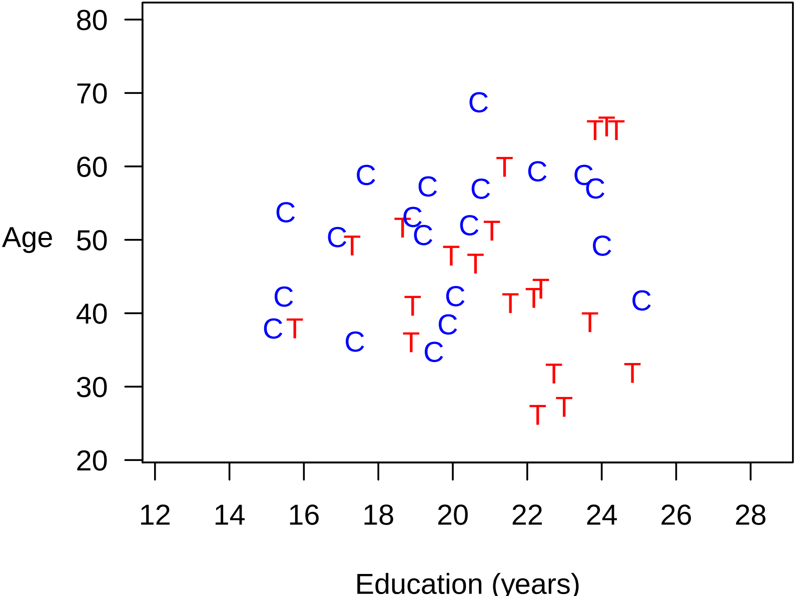

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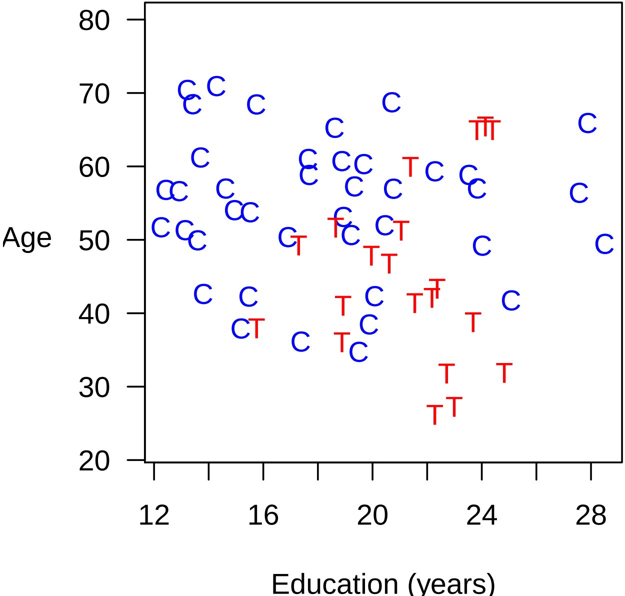

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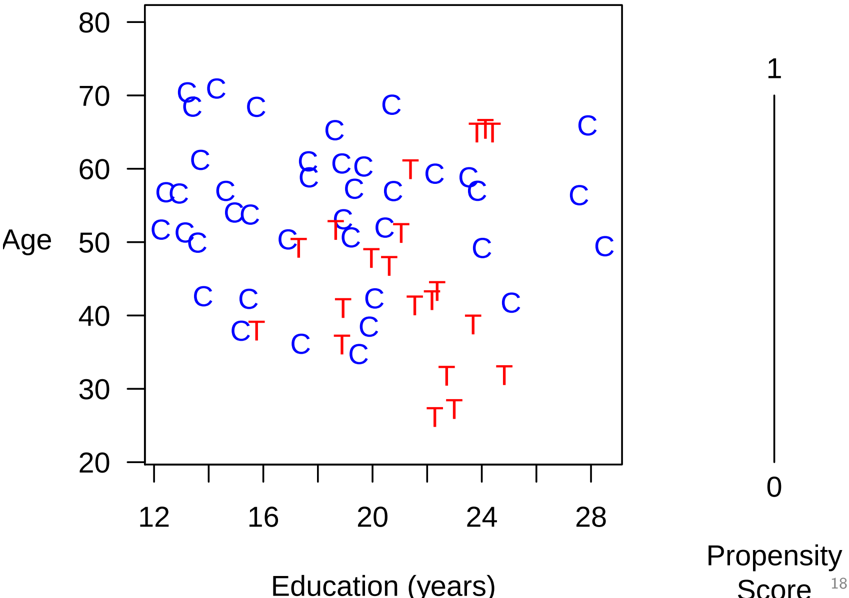

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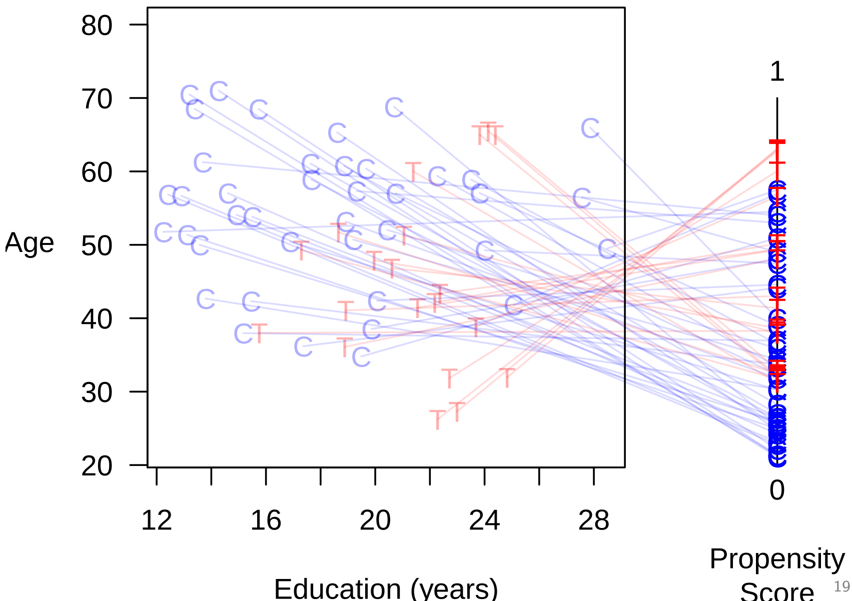

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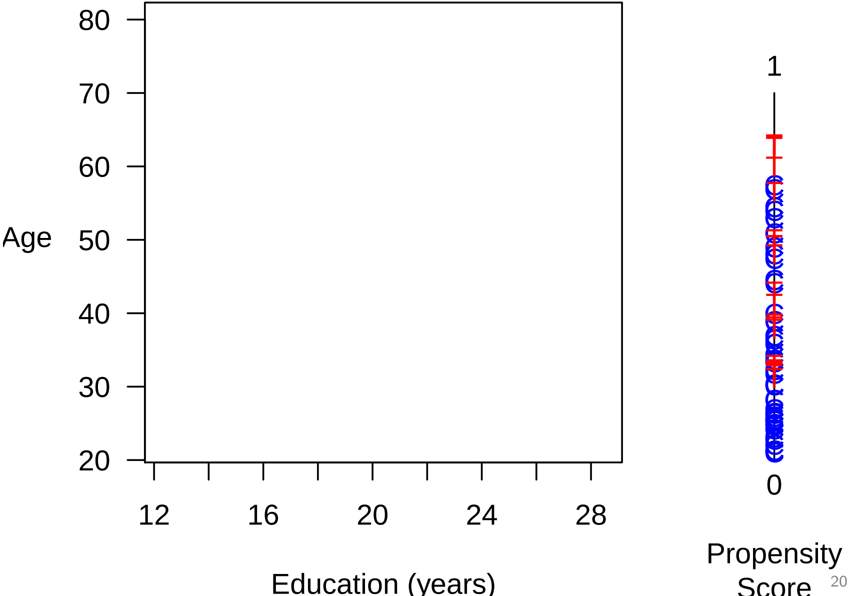

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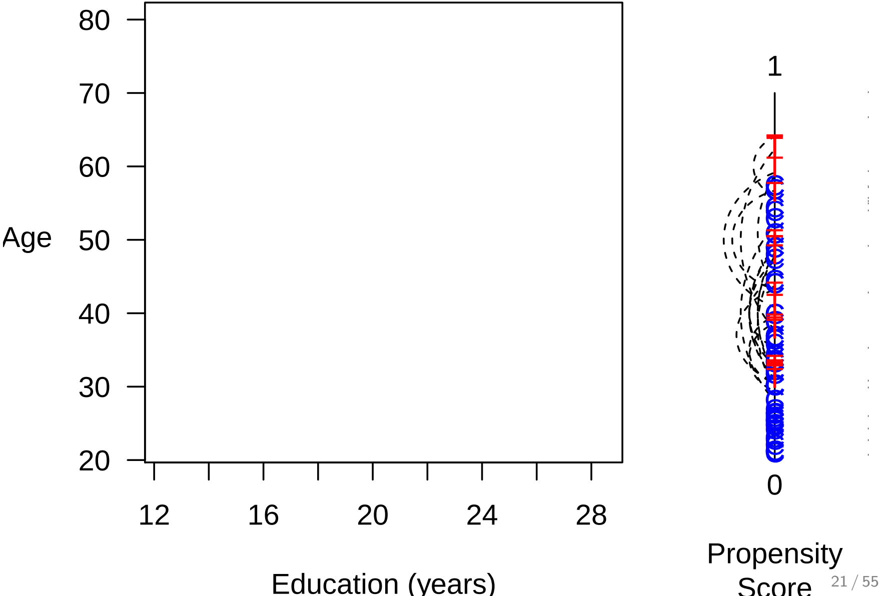

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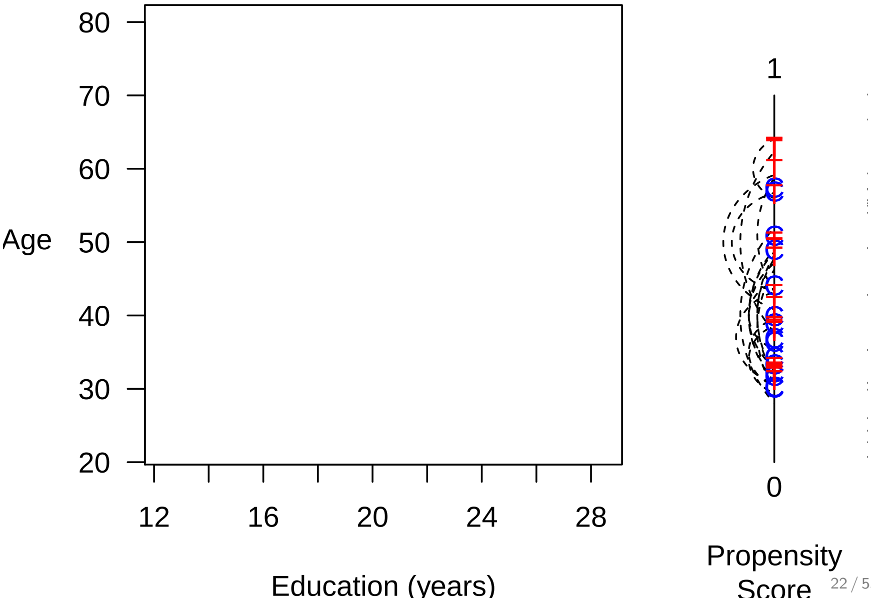

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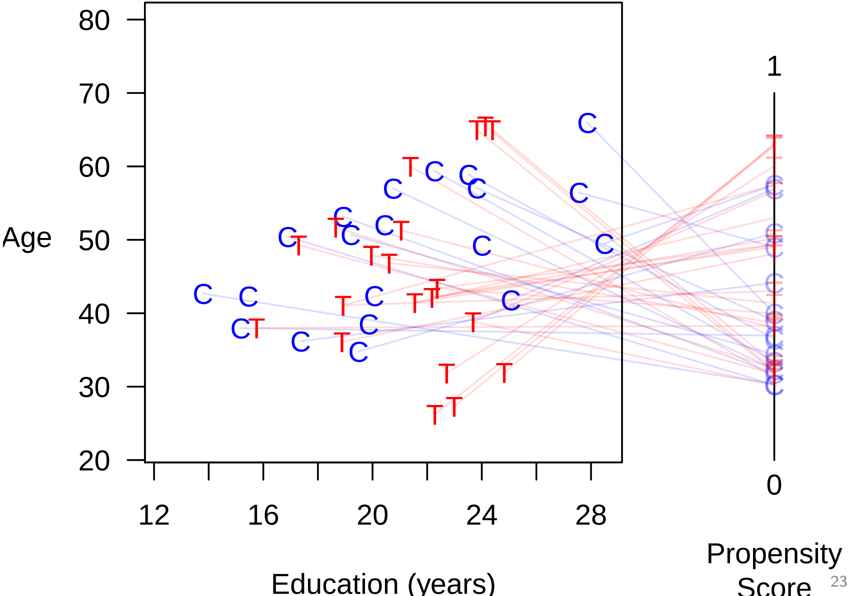

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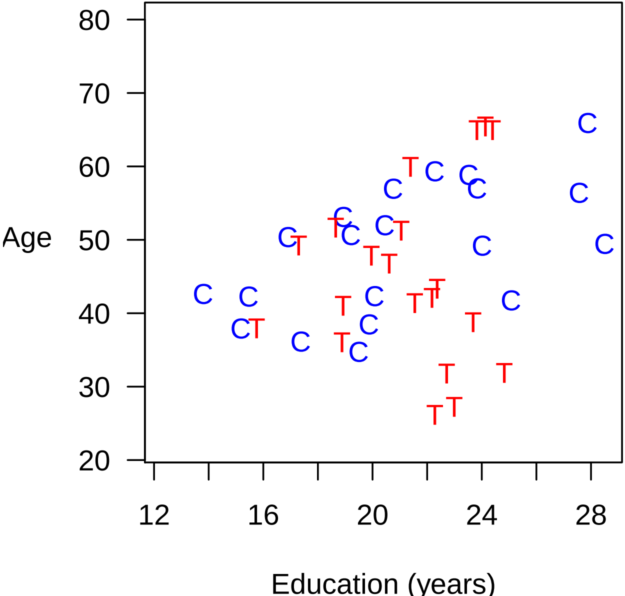

Method 3: Coarsened Exact Matching

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2. Estimation Difference in means or a model

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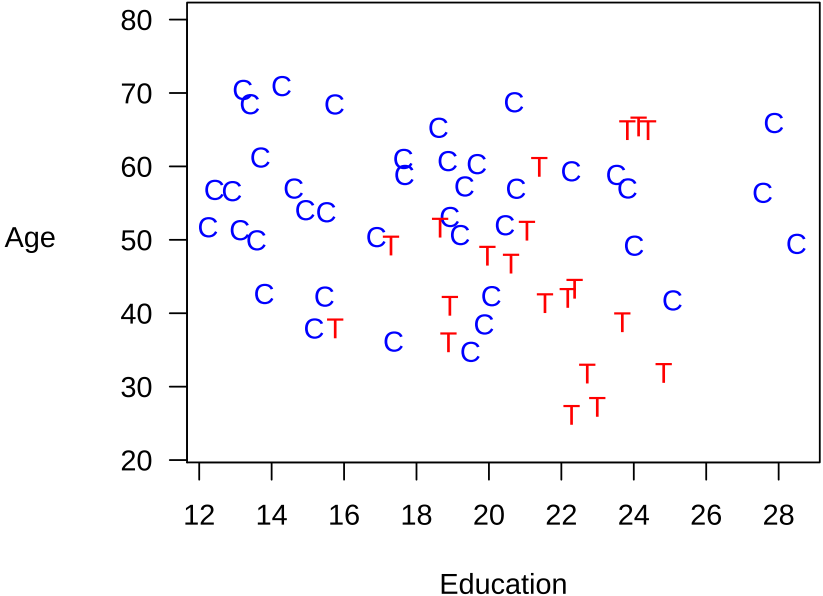

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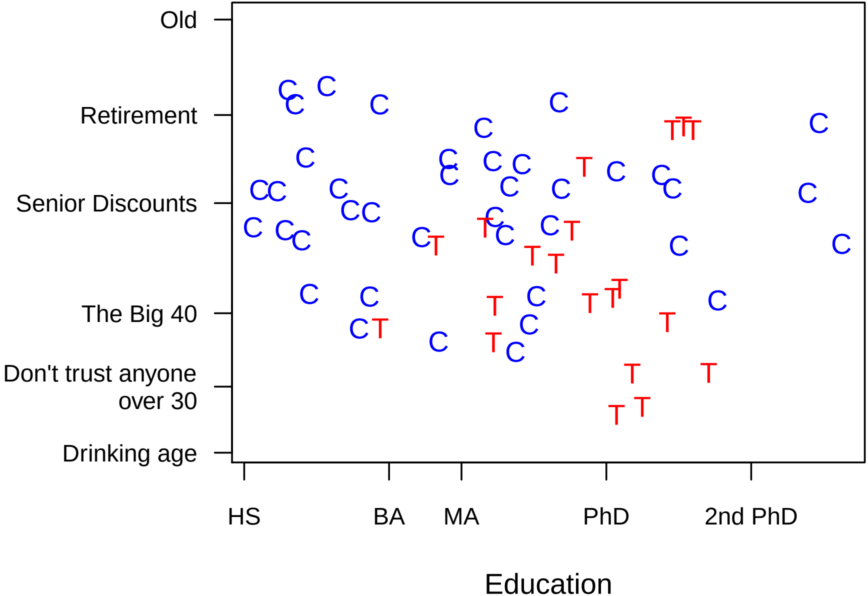

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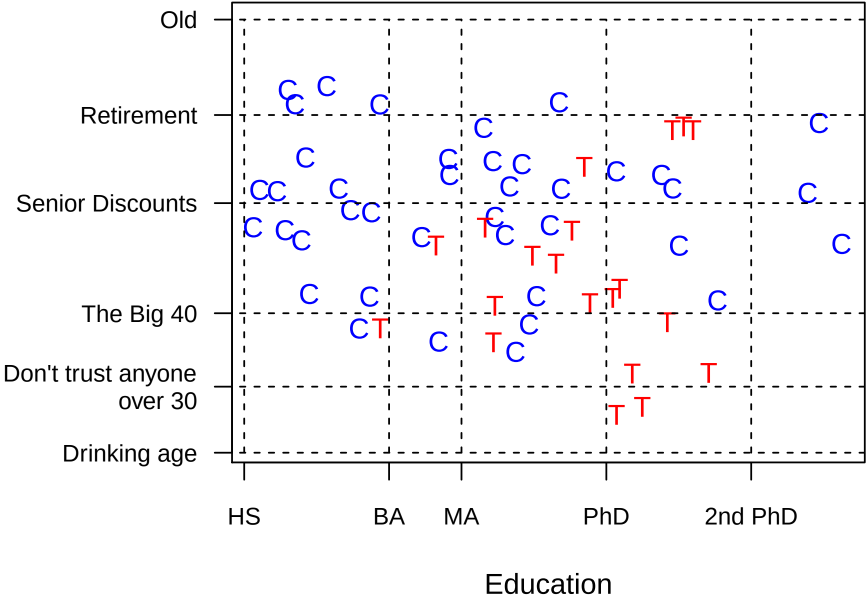

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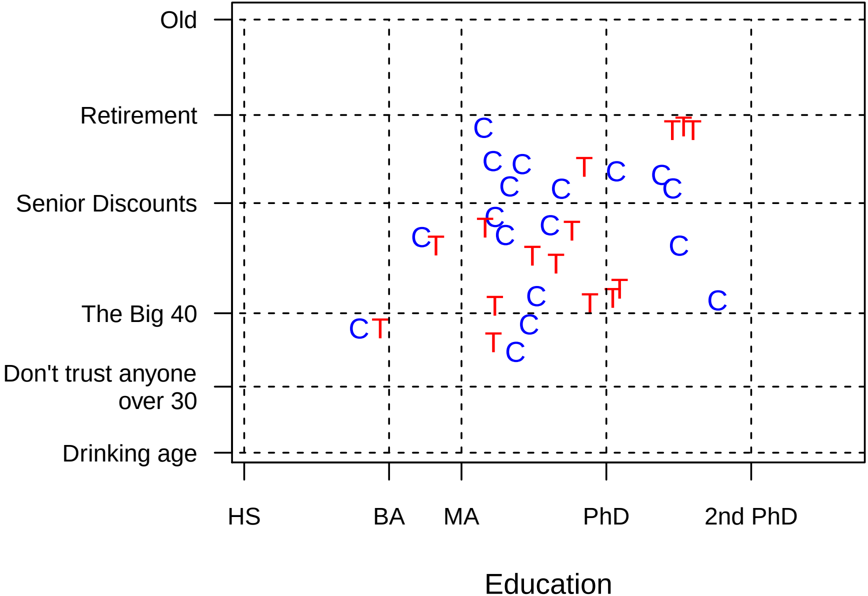

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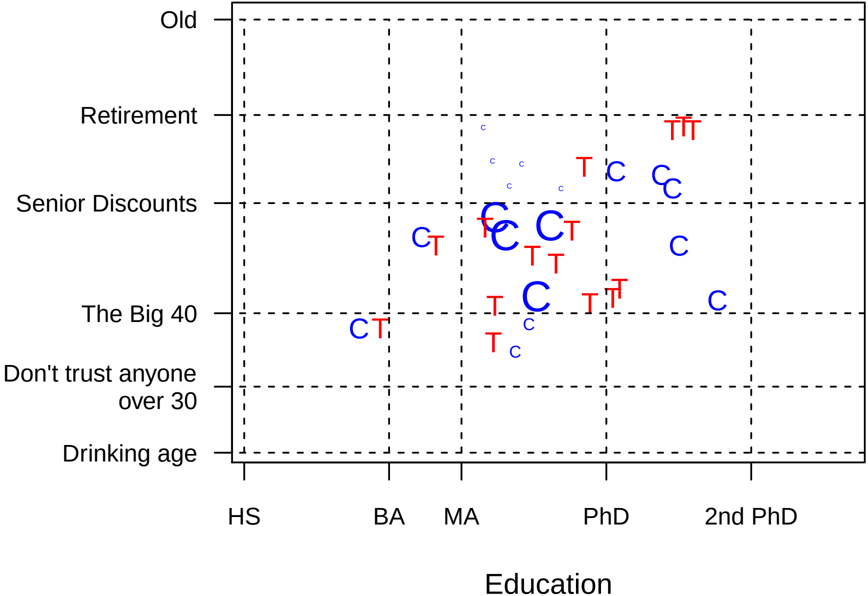

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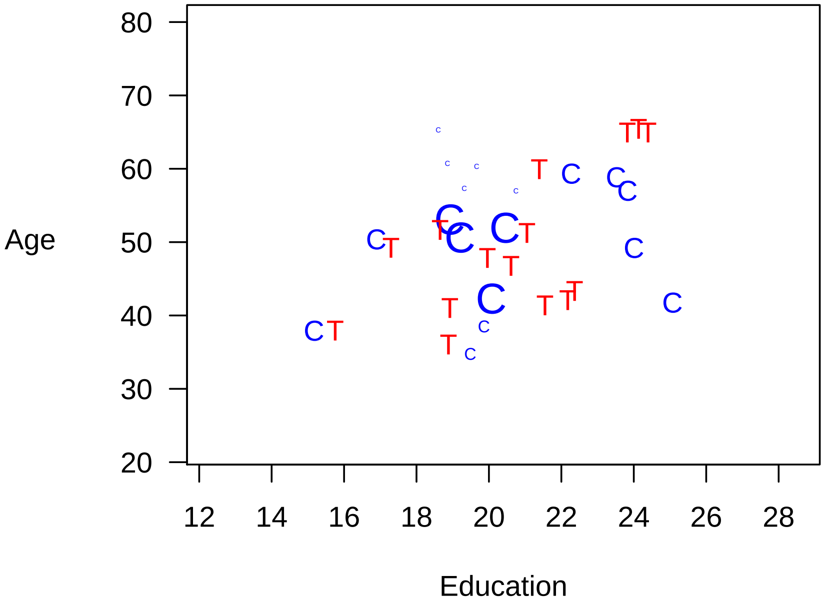

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Job Training Data: Frontier and Causal Estimates

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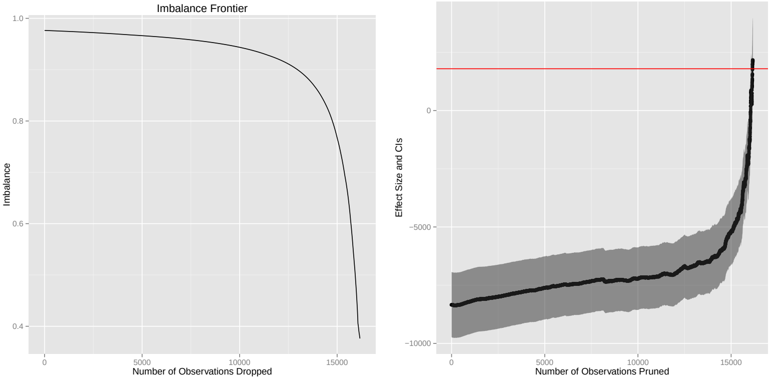

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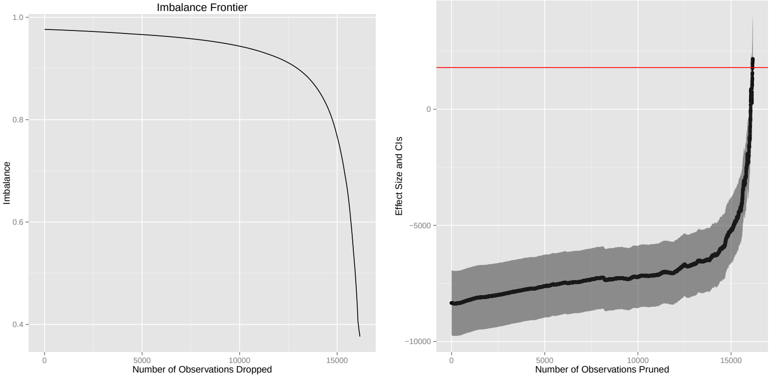

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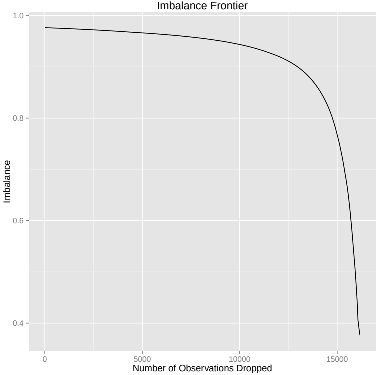

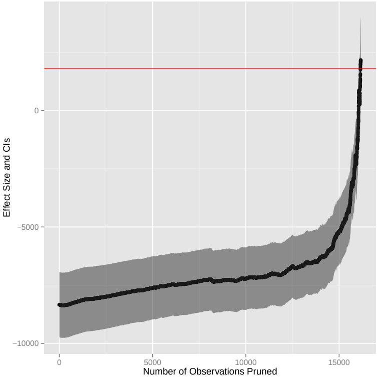

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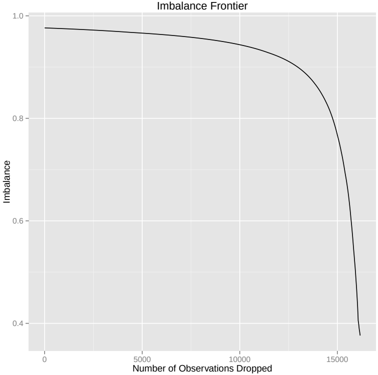

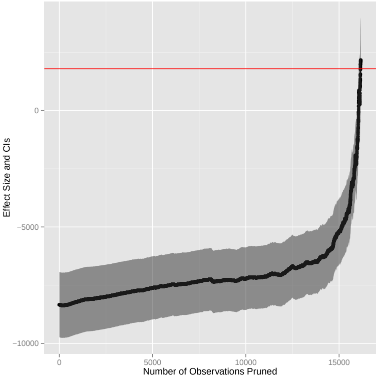

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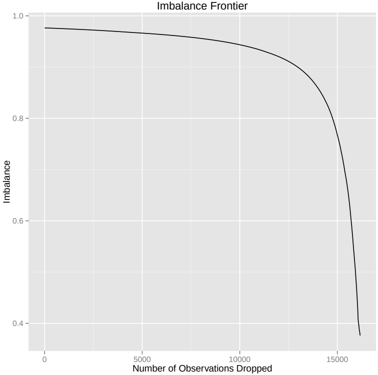

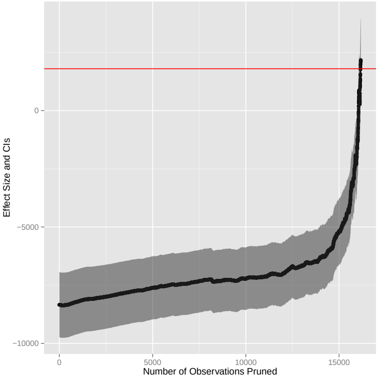

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Aid Shocks Data: Frontier and Causal Estimates

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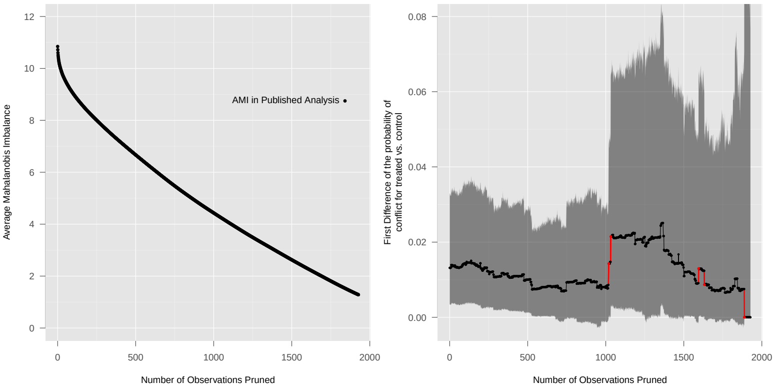

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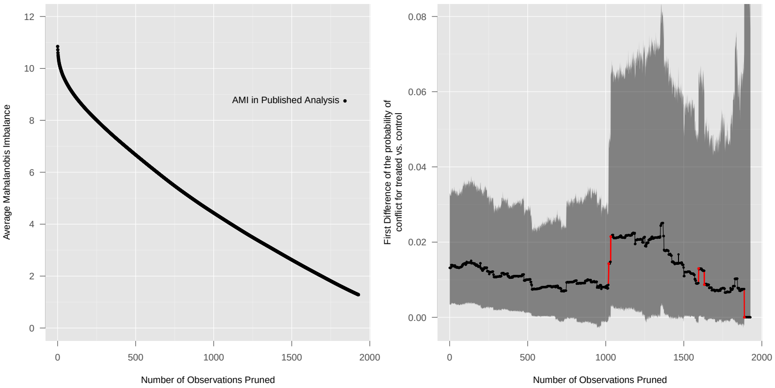

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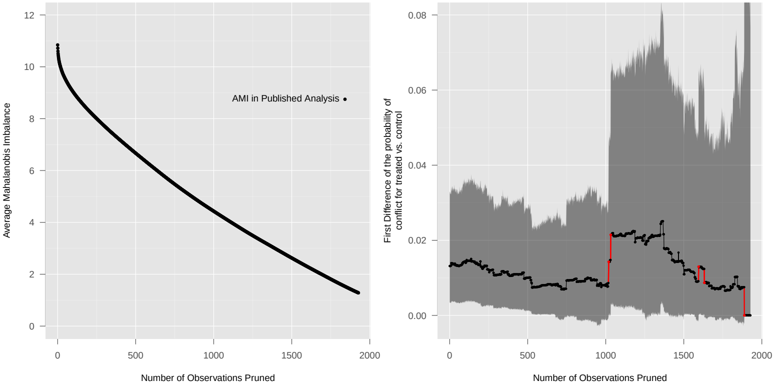

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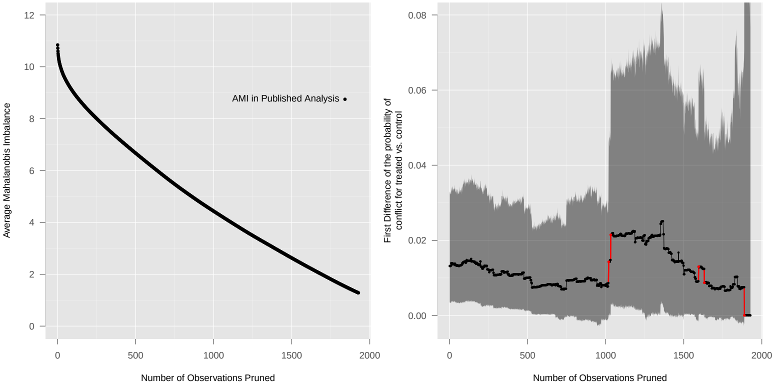

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- More difficult inferential task

Aids Shocks: Change in Quantity of Interest

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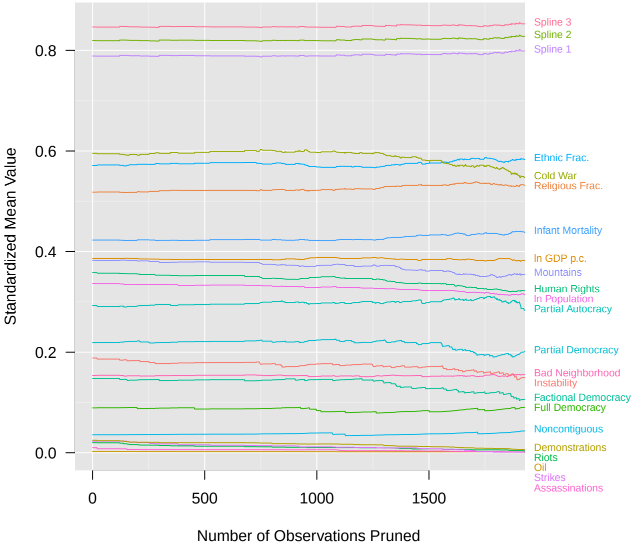

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- High leverage points
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Aids Shocks: Large Unit-Level Effects

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- High leverage points
- Cases with few substitutes
- Not model dependence (which matching helps with), but data dependence

Constructing the FSATT Mahalanobis Frontier

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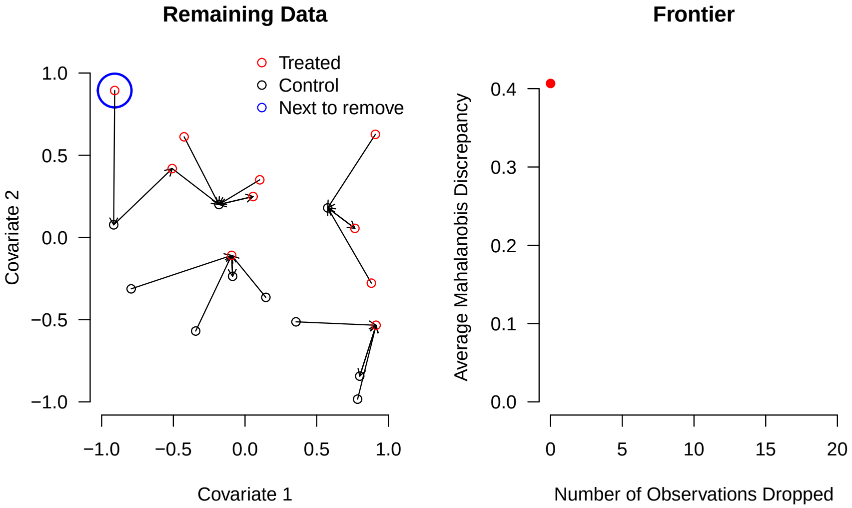

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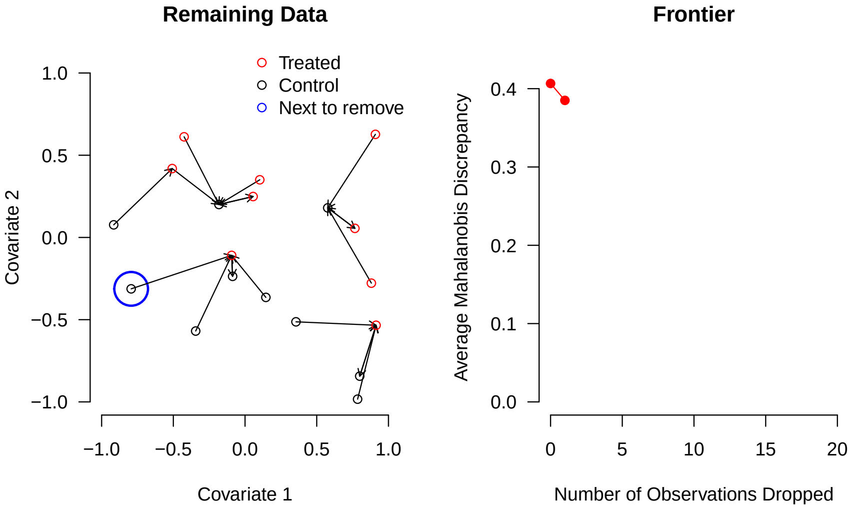

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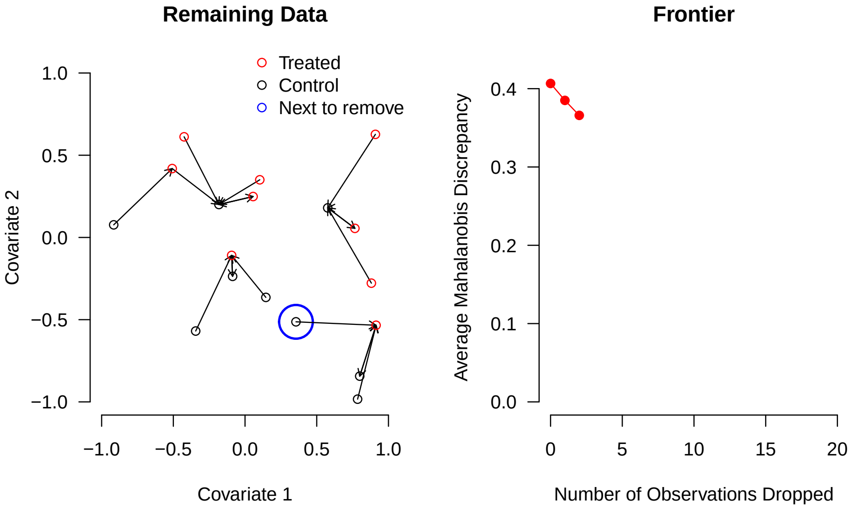

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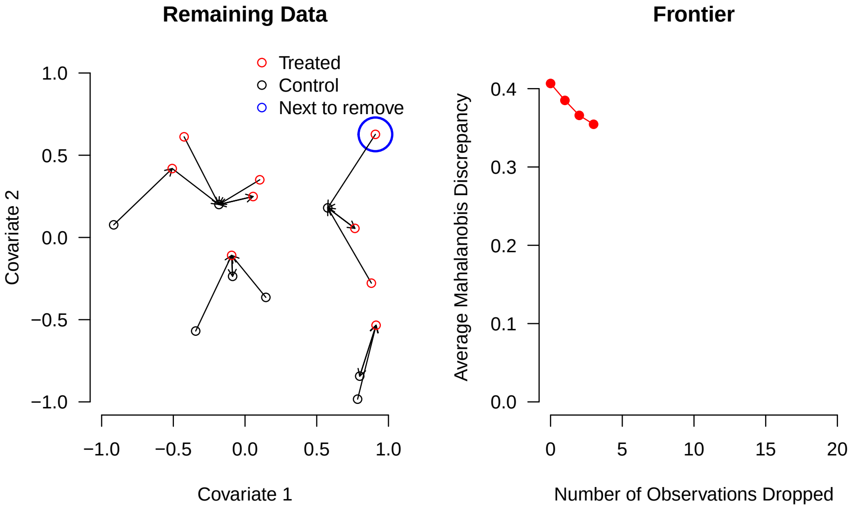

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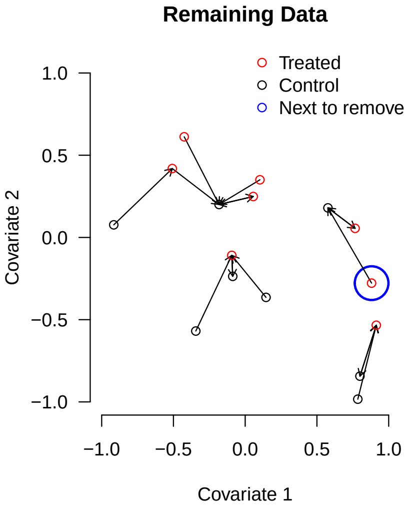

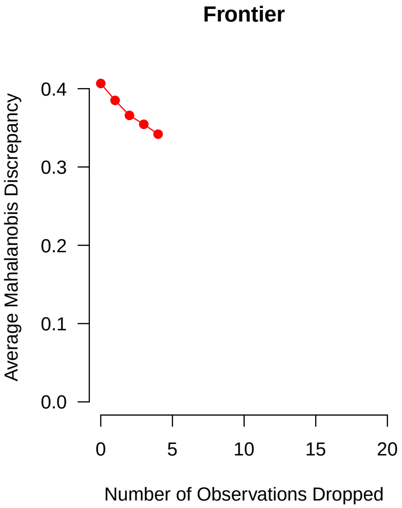

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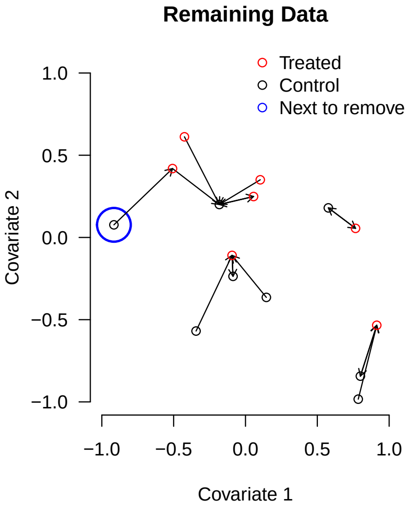

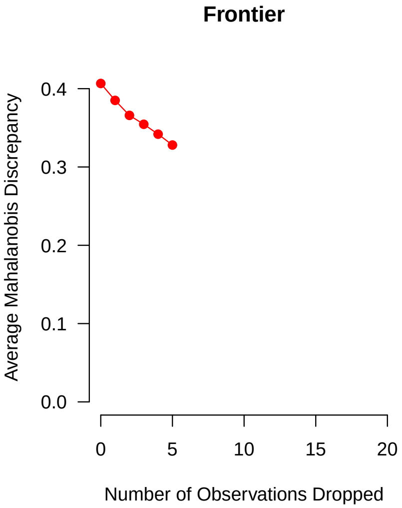

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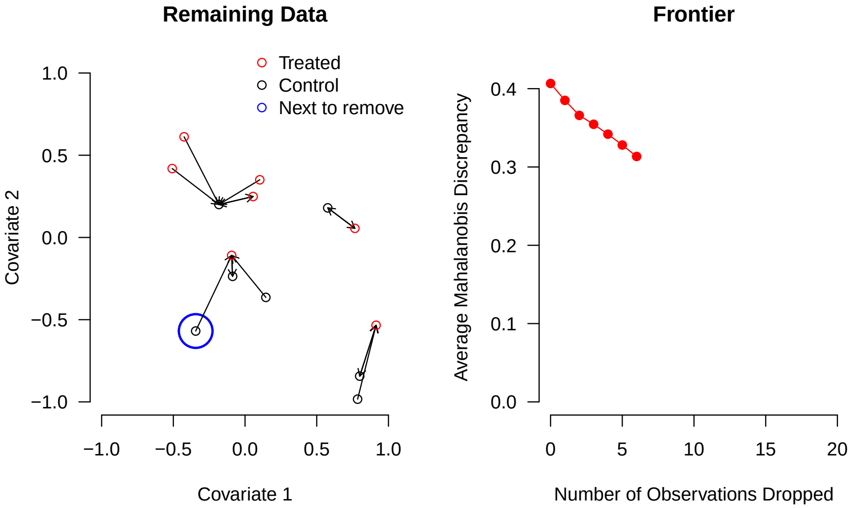

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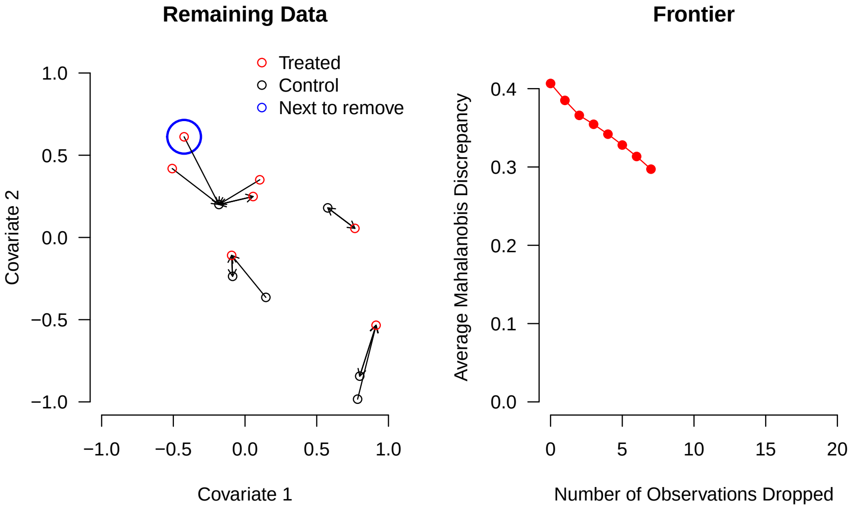

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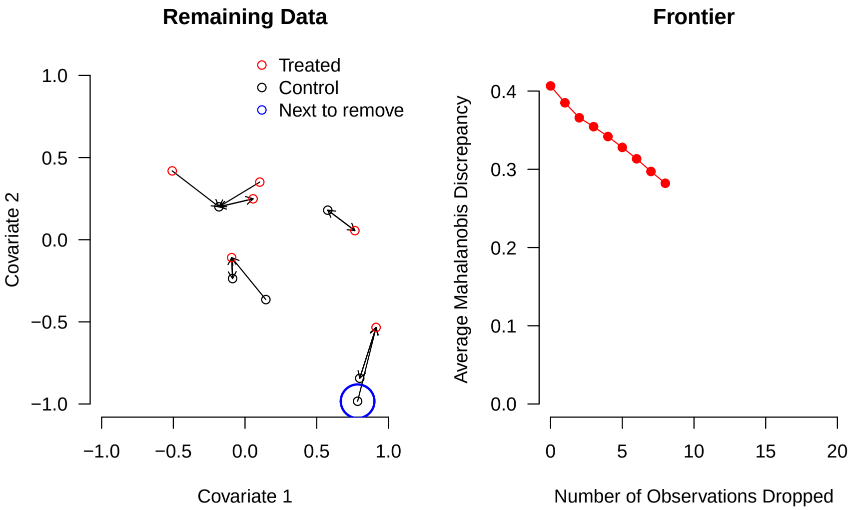

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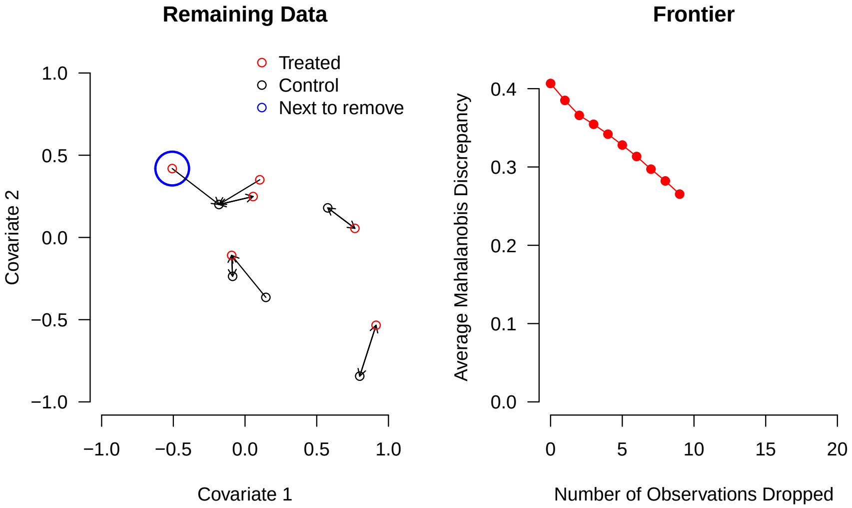

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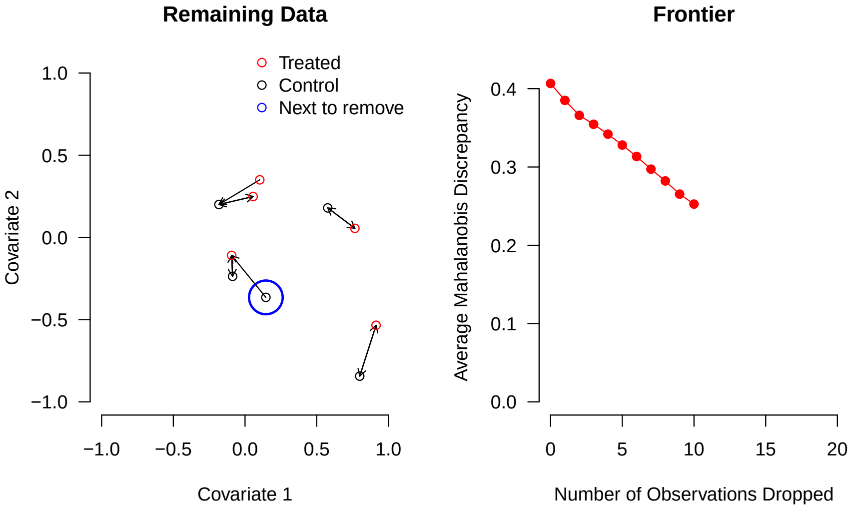

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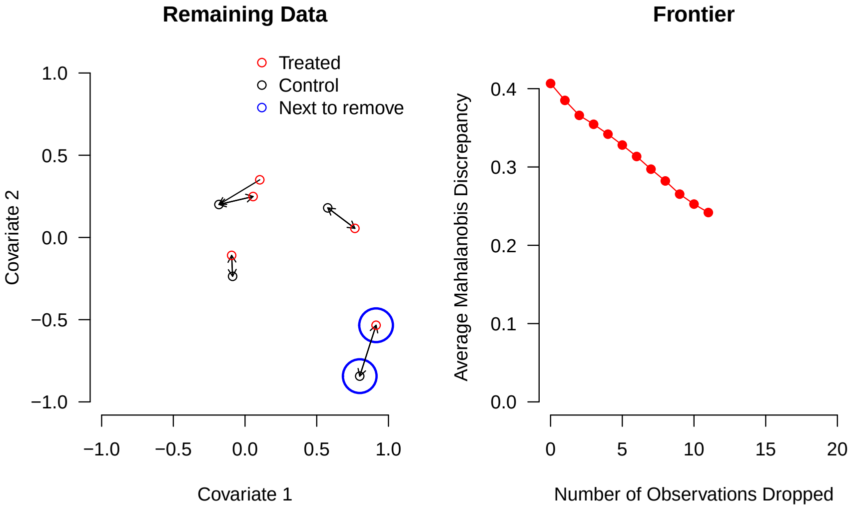

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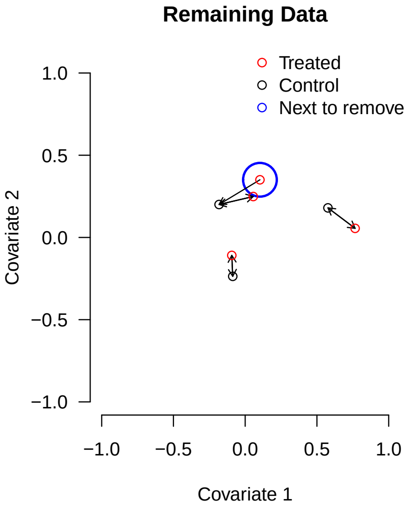

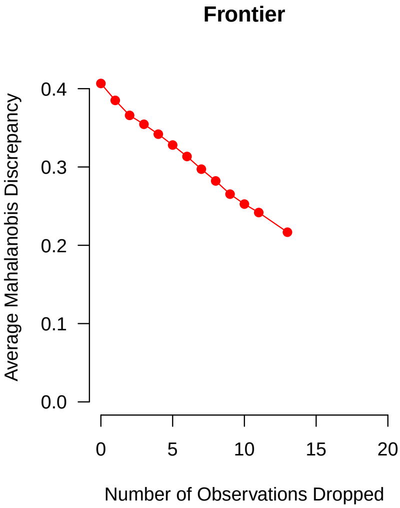

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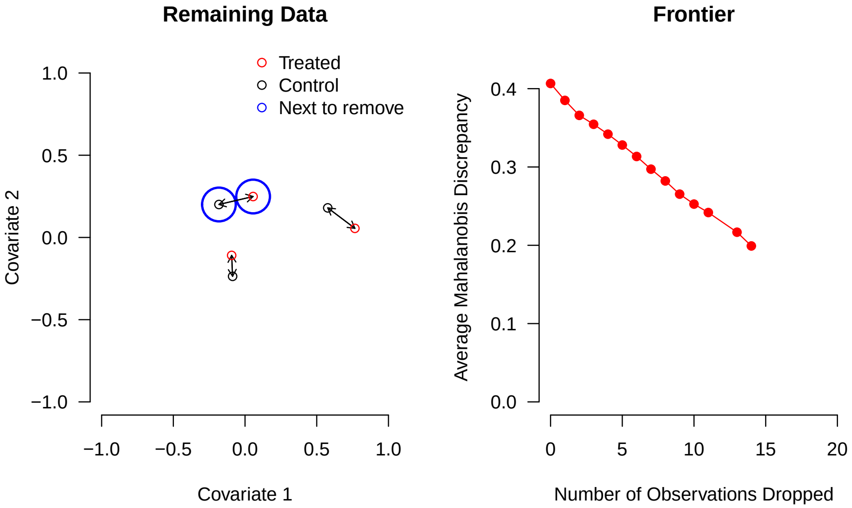

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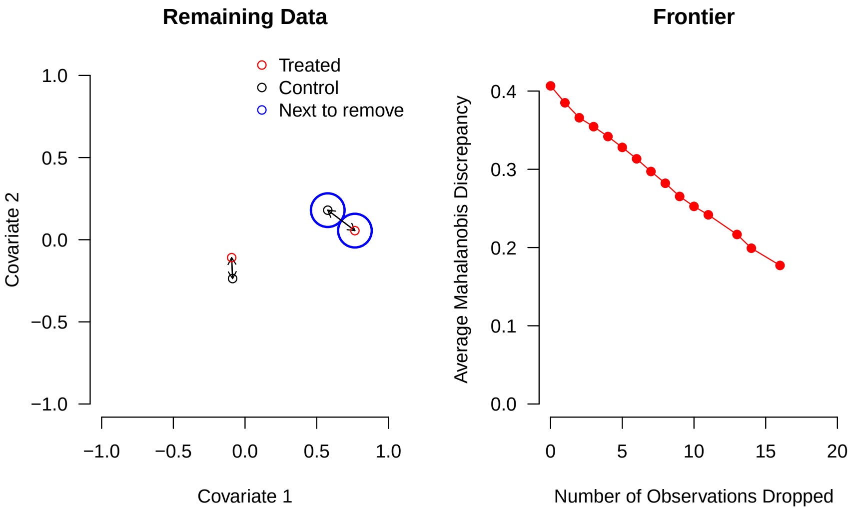

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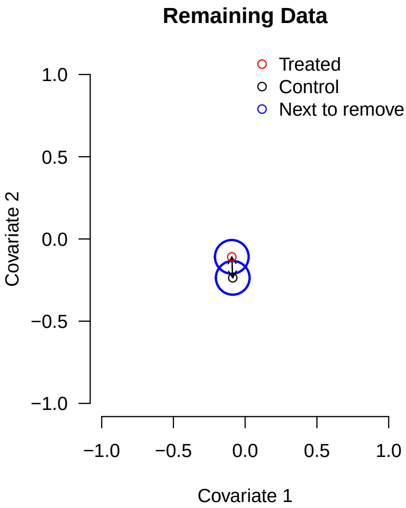

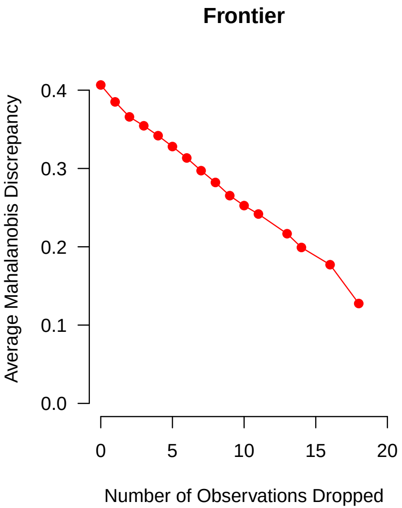

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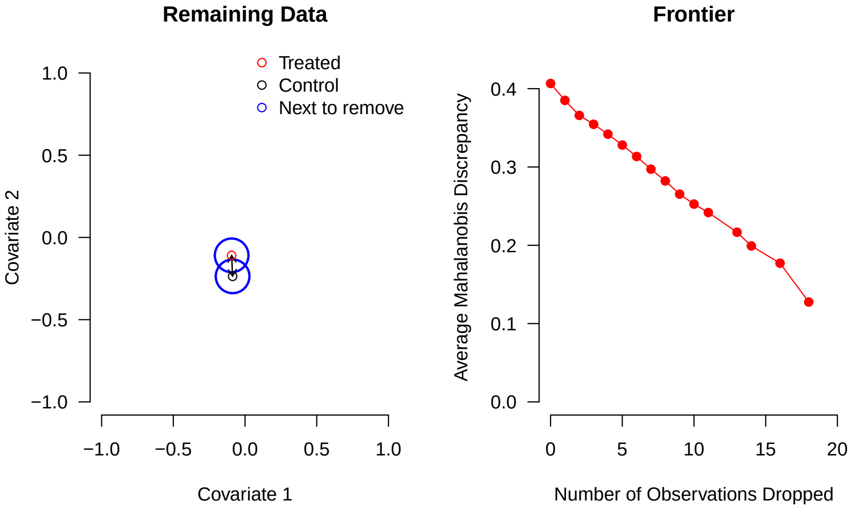

- Warning: figure omits some details!

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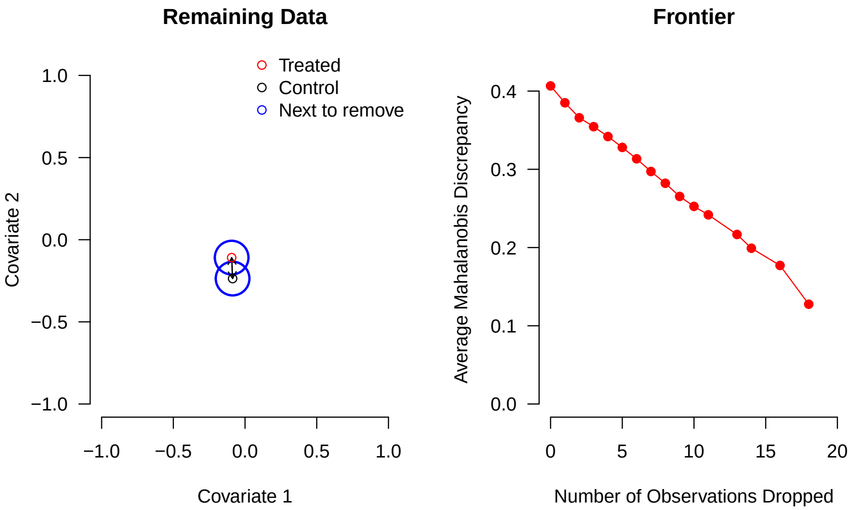
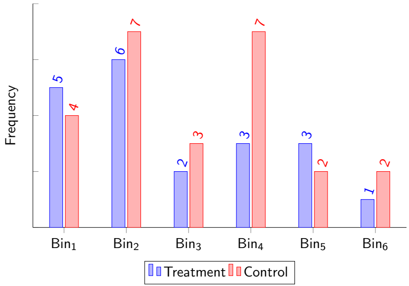
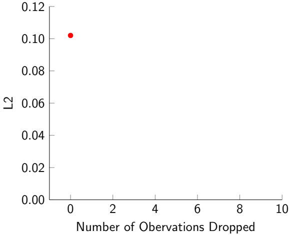


- Warning: figure omits some details!
- Very fast; works with any continuous imbalance metric

Constructing the L1/L2 SATT Frontier





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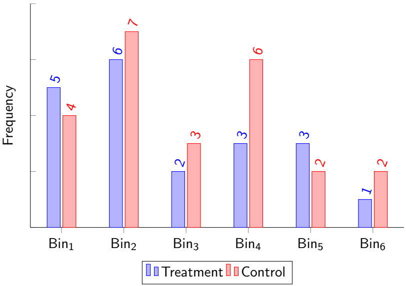

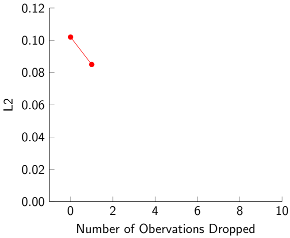

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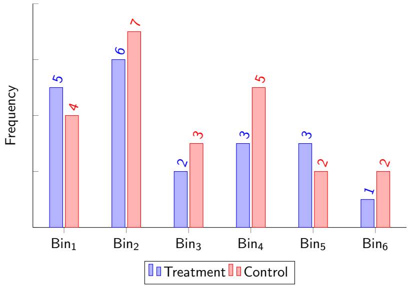

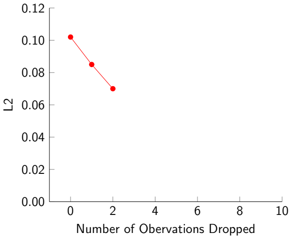

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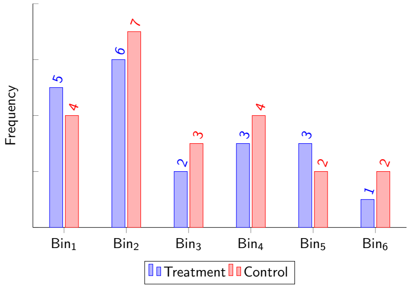

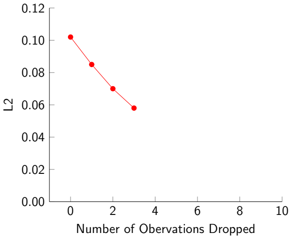

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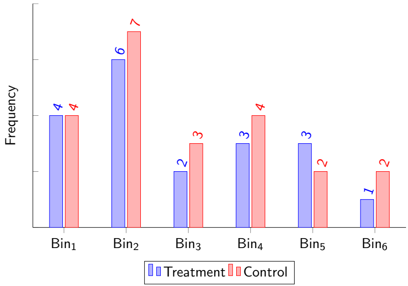

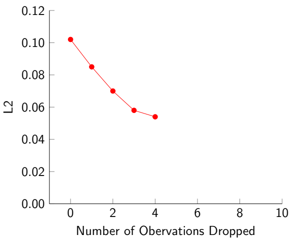

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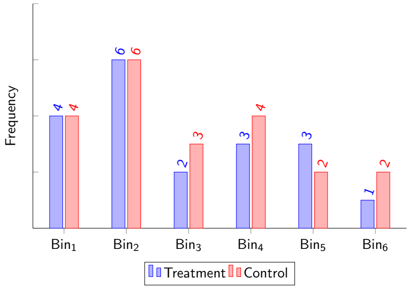

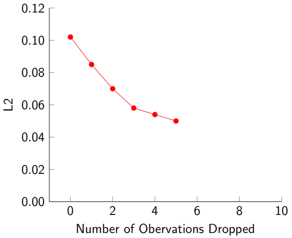

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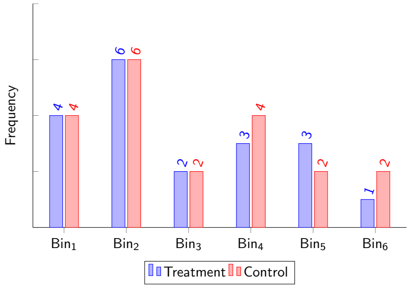

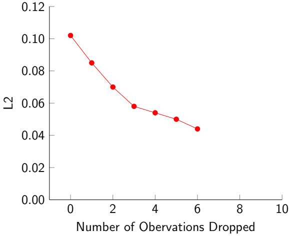

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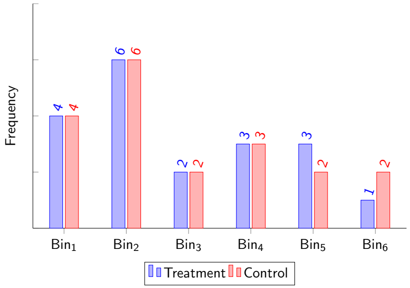

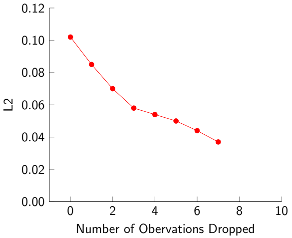

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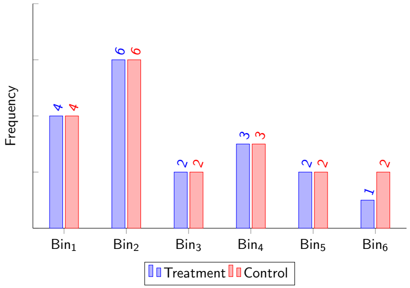

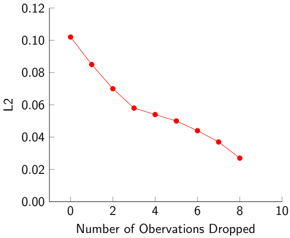

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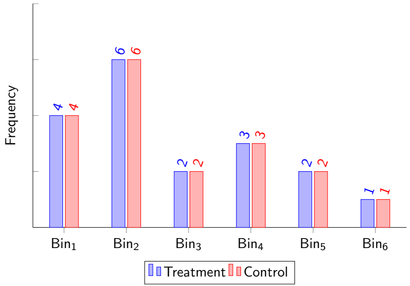

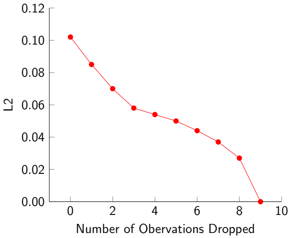

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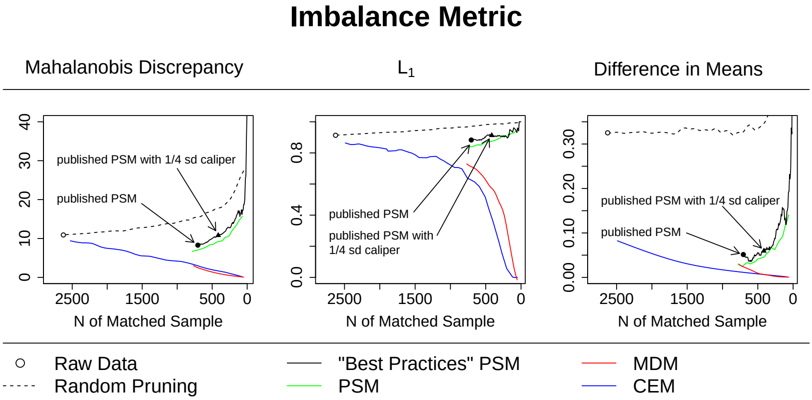
- Works very fast, even with very large data sets

Problems with PSM: Foreign Aid Shocks

King, Nielsen, Coberley, Pope, and Wells (2012)



Methods-specific frontiers (for methodological research only)

Problems with PSM: Healthways Data

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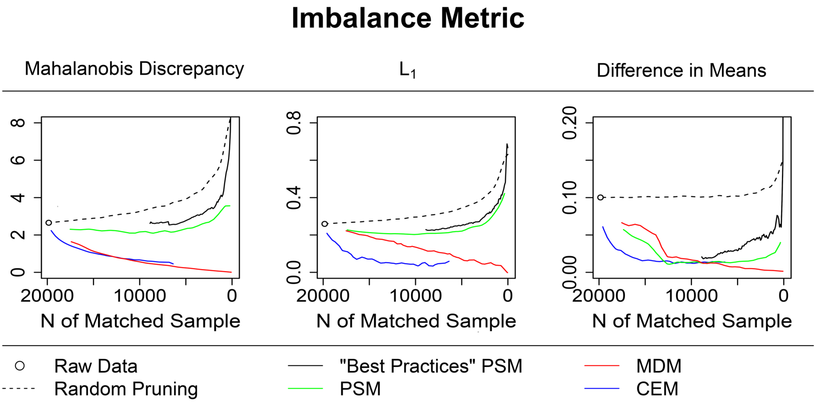

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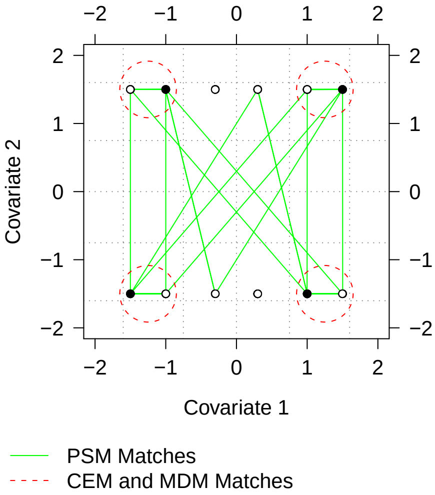
PSM Approximates Random Matching in Balanced Data



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- Software on its way · · ·

For more information

GaryKing.org